

Bioinfo Assignment 2

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Question 1

Listing 1: q1.py

```
1 import networkx as nx
2 import numpy as np
3
4
5 def main():
6     G = nx.barabasi_albert_graph(n=30, m=2, seed=42)
7
8     A = nx.to_numpy_array(G, dtype=int, nodelist=sorted(G.nodes()))
9     np.savetxt("adjacency_matrix.csv", A, fmt="%d", delimiter=",")
10
11     nx.write_graphml(G, "network.graphml")
12
13
14 if __name__ == "__main__":
15     main()
```

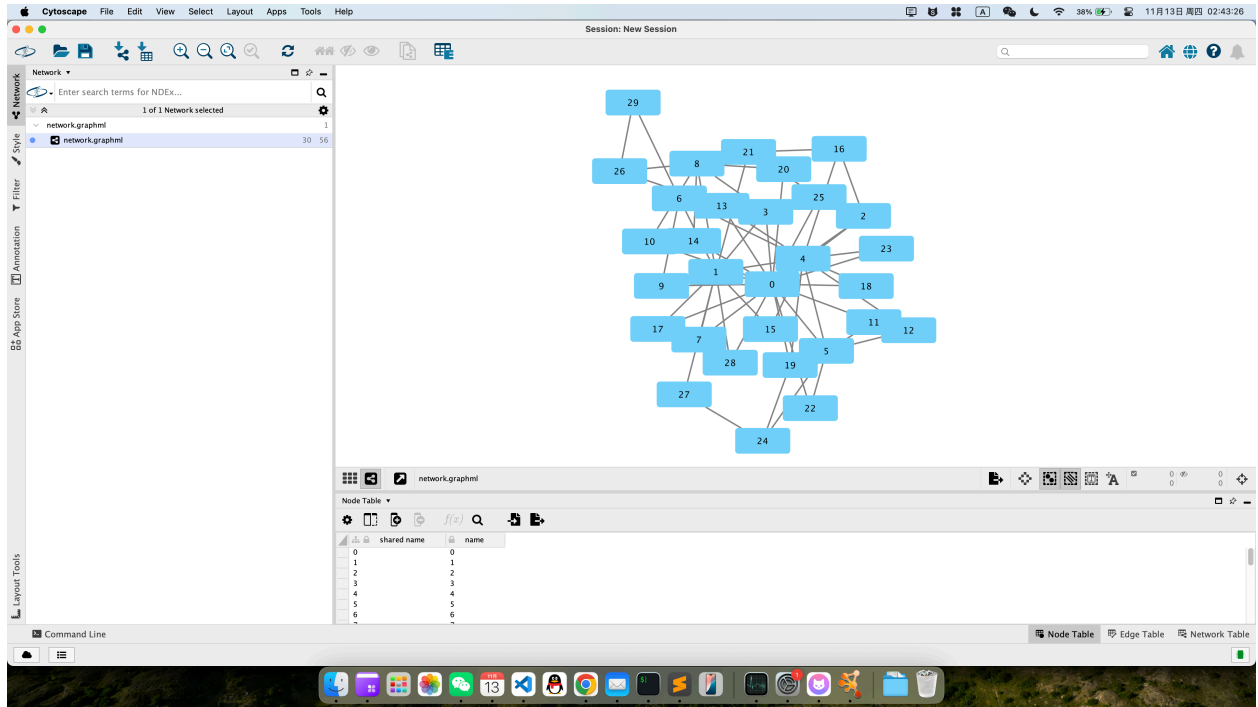


Figure 1: Cytoscape BarabásiAlbert network (n=30)

Listing 2: Adjacency Matrix (CSV format)

```

1 0,1,1,1,0,1,0,1,0,1,1,1,0,1,1,0,0,1,1,1,0,1,1,0,1,0,0,1,0
2 1,0,0,1,1,0,1,1,1,0,0,0,0,0,0,1,0,1,0,0,0,1,0,0,0,0,0,1,1,0
3 1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0
4 1,1,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
5 0,1,1,0,0,1,1,0,0,0,0,0,1,1,0,1,1,0,1,1,0,0,0,1,0,0,0,0,0,0
6 1,0,0,0,1,0,0,0,0,0,0,1,1,0,0,0,0,0,0,0,0,1,0,1,0,0,0,0,0,0
7 0,1,0,0,1,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0
8 1,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
9 0,1,0,1,0,0,0,0,0,0,1,0,0,0,1,0,0,0,0,0,1,0,0,0,0,0,1,0,0,0
10 1,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
11 1,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
12 1,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
13 0,0,0,0,1,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
14 1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0
15 1,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
16 0,1,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
17 0,0,1,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0
18 1,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
19 1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
20 1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0
21 1,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
22 0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,0,1,0,0,0
23 1,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0
24 1,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0,0

```

25	0,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,0,0,0,1,0,0
26	1,0
27	0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,0,0,0,0,1
28	0,1,0,1,0,0,0,0
29	1,1,0
30	0,0,0,0,0,0,0,1,0,1,0,0

Question 2

1. Degree of x :

$$\deg(x) = 5$$

since x is connected to nodes $\{1, 2, 3, 4, 5\}$.

2. Betweenness centrality of x :

There are $\binom{5}{2} = 10$ possible pairs of leaf nodes.

The number of shortest paths passing through x is 9.

$$B(x) = 9$$

Normalized betweenness:

$$B_{\text{norm}}(x) = \frac{9}{10} = 0.9$$

3. Characteristic path length of the network:

Total number of node pairs:

$$\binom{6}{2} = 15$$

Shortest path lengths:

distance = 1 for 6 pairs: $(x, 1), (x, 2), (x, 3), (x, 4), (x, 5), (4, 5)$

distance = 2 for 9 pairs: all other leafleaf pairs.

Hence,

$$L = \frac{6 \times 1 + 9 \times 2}{15} = \frac{24}{15} = 1.6$$

$\deg(x) = 5, \quad B(x) = 9 \ (B_{\text{norm}} = 0.9), \quad L = 1.6$
--

Question 3

Listing 3: q3.py

```
1 import numpy as np
2 import pandas as pd
3
4 genes = ["G1", "G2", "G3", "G4"]
5 MI = pd.DataFrame([
6     [1.24, 1.01, 0.69, 0.69],
7     [1.01, 1.56, 1.01, 0.78],
8     [0.69, 1.01, 1.01, 0.46],
9     [0.69, 0.78, 0.46, 1.01]
10 ], index=genes, columns=genes)
11
12 # calc Zi
13 Zi = MI.apply(lambda x: (x - x.mean()) / x.std(ddof=0), axis=1)
14 Zi = Zi.clip(lower=0)
15
16 # calc Zj
17 Zj = MI.apply(lambda x: (x - x.mean()) / x.std(ddof=0), axis=0)
18 Zj = Zj.clip(lower=0)
19
20 # calc CLR mat
21 CLR = np.sqrt(Zi ** 2 + Zj ** 2)
22 CLR = pd.DataFrame(CLR, index=genes, columns=genes)
23
24 pairs = []
25 for i in range(len(genes)):
26     for j in range(i + 1, len(genes)):
27         pairs.append((genes[i], genes[j], CLR.iloc[i, j]))
28
29 top3 = sorted(pairs, key=lambda x: x[2], reverse=True)[:3]
30
31 print("Zi_mat\n", Zi.round(3))
32 print("\nZj_mat\n", Zj.round(3))
33 print("\nCLR_mat\n", CLR.round(3))
34 print("\nTop_3_GGI")
35 for g1, g2, val in top3:
36     print(f"{g1}-{g2}: {val:.3f}")
```

Row-wise positive z -scores (Z_i):

$$Z_i = \begin{bmatrix} 1.432 & 0.441 & 0 & 0 \\ 0 & 1.637 & 0 & 0 \\ 0 & 0.937 & 0.937 & 0 \\ 0 & 0.228 & 0 & 1.396 \end{bmatrix}.$$

Column-wise positive z -scores (Z_j):

$$Z_j = \begin{bmatrix} 1.432 & 0 & 0 & 0 \\ 0.441 & 1.637 & 0.937 & 0.228 \\ 0 & 0 & 0.937 & 0 \\ 0 & 0 & 0 & 1.396 \end{bmatrix}.$$

CLR matrix:

$$CLR = \begin{bmatrix} 2.025 & 0.441 & 0 & 0 \\ 0.441 & 2.315 & 0.937 & 0.228 \\ 0 & 0.937 & 1.325 & 0 \\ 0 & 0.228 & 0 & 1.974 \end{bmatrix}.$$

Top 3 GGI

1. $(G2, G3)$: $CLR_{23} = 0.937$,
2. $(G1, G2)$: $CLR_{12} = 0.441$,
3. $(G2, G4)$: $CLR_{24} = 0.228$.

Question 4

(a) What is the major mutation signature contribute to AML?

According to the given matrix, AML is associated with **Signature 1** and **Signature 5**.

Signature 1 is the most dominant across all cancer types and typically shows stronger contribution in AML.

(b) What is the major mutation type related to AML?

C>T is accounted for more than 10% of all.

(c) Signatures Dominated by T>A

Signature 22 and Signature 27 are dominated by T>A, which is more than 10%.

Associated Cancer Types: Liver cancer, Urothelial carcinoma (Signature 22); Kidney clear cell carcinoma (Signature 27).

(d) Model Design

Feature Representation

Each patient can be represented by:

- **Raw mutation spectrum** (96-dimensional trinucleotide mutation frequencies), or
- **Signature contributions** obtained from NMF (H vector).

Model Choice

As far as the task is concerned, we can use a two-step model:

NMF decomposition $\Rightarrow$ Random Forest classifier

Procedure

1. Build matrix X from patient mutation profiles.
2. Apply NMF to obtain H (signature weights per patient).
3. Split the data into training and test sets.
4. Train a Random Forest or Logistic Regression model using H as input and cancer type as target.
5. For a new patient, compute mutation profile $\rightarrow$ project to H space $\rightarrow$ predict cancer type.

Why This Model Fits the Task

- NMF provides biologically interpretable latent features (mutation signatures).
- The classifier captures statistical relationships between signature patterns and cancer types.
- The model can generalize to unseen patients with similar mutational patterns.
- Probability outputs allow risk estimation and uncertainty quantification.